

Title: Exploring Long COVID Outcomes in the United States: Prevalence, Limitations, and Racial Disparities

i. Introduction & Motivation

Long COVID has a silent impact on numerous people in the United States. It is a major public health concern and impacts quality of life to varying degrees. As a first-hand individual who experiences Long COVID, it is important that I explore and learn about its patterns. This project examines in greater depth the patterns in prevalence, limitations, and racial disparities among affected patients. The dataset chosen for this project was from data.gov titled *Post-Covid Conditions*. The U.S. Department of Health and Human Services began this study from April 23, 2020, to April 23, 2025, spanning five years of data.

The dataset contains 18,639 rows and 16 columns. There are survey-based estimates reporting the prevalence of COVID infection history and Long COVID experiences. Additionally, there are subgroups including sex, age, race/ethnicity, disability status, education level, and state. The dataset is divided into time periods within the five-year window.

The three questions I am looking to discover during my Exploratory Data Analysis include:

- 1. How do national COVID and Long COVID prevalence measures compare?**
- 2. How do activity limitations differ between adults with Long COVID and the overall adult population?**
- 3. Do Long COVID rates differ by race/ethnicity?**

ii. Data Understanding & Preparation

There is a combination of categorical and numerical key values in this dataset. The key categorical variables include *Indicator* (a description of those who are currently having symptoms of Long COVID or activity limitations, etc.), *Group* (describing the demographic category), and *Subgroup* (a variable that groups similar data, e.g., ages 30-39, sex: female, race: Hispanic). The key numerical variables include *Value* (the reported percentage of each indicator and subgroup), as well as *LowCI* and *HighCI* (the lower and upper bounds of the confidence intervals). The dataset also includes time variables, which are *Time Period*, *Start Date*, and *End Date* (representing how long the collection for that session lasted and the days it began and ended, allowing for comparison amongst time periods).

To ensure clean data, various preprocessing steps were executed. These steps include standardizing column names (e.g., removing trailing or leading whitespaces), converting the *Value* column into a numeric format, converting date fields into a datetime format, and including responses from the most recent study only. Missing values were evaluated by percentage calculations and absolute counts. The indicators *Value*, *LowCI*, *HighCI*, and *Confidence Interval* all contained approximately 13% of missing values. Since this project primarily focuses on the information provided from the *Value* variable, any rows with that missing value were excluded using `.dropna()`, which was applied during plotting. The missing values were left unfilled as it describes potential reporting limitations rather than errors.

Data quality considerations within this dataset include the use of different denominators across indicators. For example, examining adults who have Long COVID

is calculated from the entire adult population, whereas examining adults with activity limitations from Long COVID is taken from the Long COVID subgroup. Therefore, percentage results are not directly comparable, depending on the questions. Additionally, these estimates are from self-reported data, which can provide bias in certain areas that can impact results.

iii. Exploratory Analysis

To analyze the distribution of the data across the dataset, the variable *Value* was examined. As seen in Plot 1, the data are right-skewed with a majority of the values clustered below 30%. This suggests that COVID-related indicators affect a small percentage of the overall population. However, towards the higher percentage values, there are distinct clusters. This indicates that, for those who do have Long COVID, a greater percentage of this subgroup is affected by conditional indicators due to differences in denominators (e.g., all adults vs adults with Long COVID).

In Plot 2, a boxplot was used to view the reported percentage values. While most estimates fall below 30%, shown by the median at 14% and Q3 at approximately 30%, a cluster of higher values above 70% appears as statistical outliers. These high values are not data errors but correspond to indicators measured among specific subpopulations (e.g., individuals currently experiencing Long COVID), where percentages are naturally higher due to different denominators. It is imperative to understand how the data is distributed to ensure correct comparisons between subgroups and indicators.

The national prevalence comparison of COVID and Long COVID is presented in Figure 1. This figure was chosen to examine the percentage of individuals ranging from

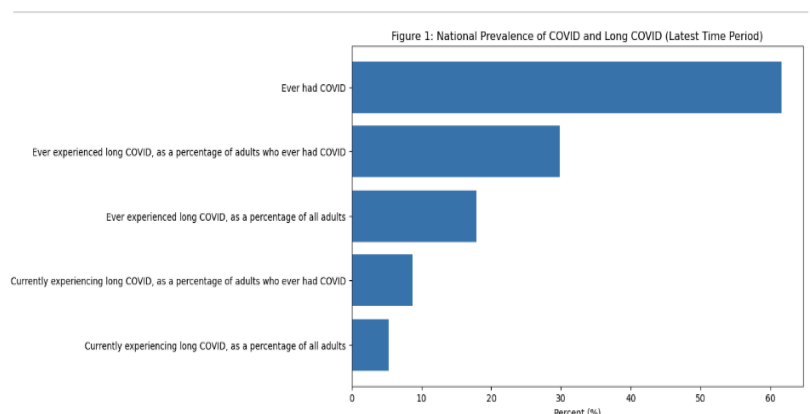
those who have had COVID to those currently experiencing Long COVID. The results display that a majority of adults had COVID, compared to those with progression into Long COVID.

There is a multivariate visualization illustrated in Figure 2 with the purpose of exploring the relationship between populations and limitations. This compares activity limitations across two populations: *all adults and adults currently with Long COVID*. Viewing both populations together highlights the vast difference in limitations.

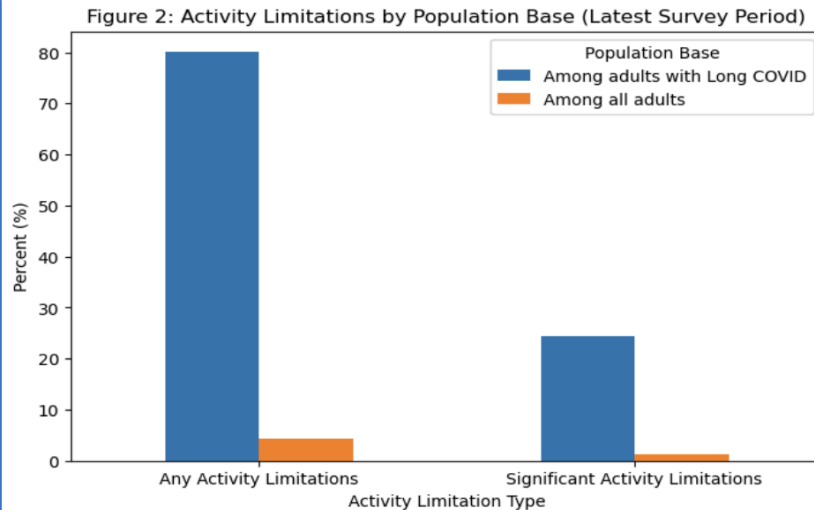
Figure 3 provides a grouped comparison of race/ethnicity by prevalence, and compares the percentage values of those currently experiencing Long COVID amongst those subgroups. This comparison was conducted to identify patterns across racial/ethnic groups and understand the variation at a more in-depth level. The results report observable differences.

4. Key Findings

Insight 1: As seen in Figure 1, a majority of adults report ever having COVID (roughly 60%), a smaller percentage report ever experiencing Long COVID (roughly 18%), and the lowest percentage report currently experiencing Long COVID (roughly 6%). This pattern suggests that although roughly 60% of people are reported of having had COVID in this study, Long COVID affects a smaller subset of



individuals. Persistent symptoms (Long COVID) are even less common than prior infection or prior Long COVID experience.

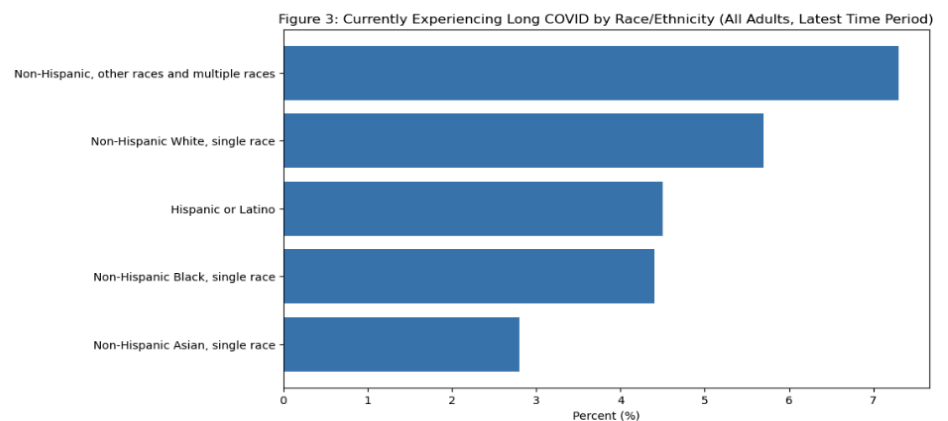


Insight 2: Figure 2 provides a clear depiction that Long COVID prevalence remains low for the overall adult population, but those who do have Long COVID are impacted by varying levels of activity limitations. About 80% of individuals

with Long COVID have *Any Activity Limitations*, and about 22% have *Significant Activity Limitations*.

Insight 3: It was observed in Figure 3 that Long COVID prevalence differs across racial and ethnic groups in the most recent survey period.

The highest reported percentage appears among individuals identifying as Non-Hispanic, other races and multiple races (7.3%), while the lowest appears among Non-Hispanic Asian adults (2.8%). Although the overall prevalence remains below 10% for all groups, the



difference between the highest and lowest groups is about 4%, indicating a noticeable gap.

5. Limitations & Next Steps

There are meaningful limitations seen throughout this dataset. The data values were self-reported, allowing for the possibility of bias. Some individuals may interpret their symptoms differently, which may lead to fluctuating meanings of symptoms.

Additionally, since the data compared different populations that had different denominators, it can be difficult when comparisons are made between them. There were also missing values within the dataset, which could potentially impact the results.

To improve future analysis, the dataset should include socioeconomic status and the severity of the infections. I believe that these indicators would provide insightful comparisons to allow for meaningful change in the healthcare sector.

This exploratory data analysis was created to make observations between indicators. There are no causal claims being made. Therefore, the patterns found within this dataset do not apply to cause-and-effect relationships.

6. Conclusion

Overall, this exploratory data analysis found that Long COVID affects a small subset of those who had COVID. While the prevalence of Long COVID is lower, those who have it are impacted by limitations. The races/ethnicities impacted most are Non-Hispanic, other races and multiple races, and the least are Non-Hispanic Asian adults. Although Long COVID affects a smaller portion of the population, the impacts are meaningful and deserve proper acknowledgement for future research.

Reference

Publisher Centers for Disease Control and Prevention. (2025a, April 23). *U.S. Department of Health & Human Services - Post-COVID Conditions*. Catalog. <https://catalog.data.gov/dataset/post-covid-conditions-89bb3>